

## EXP 5

### Calculation of Group velocity, Phase velocity and phase constant of EM waves in rectangular waveguides

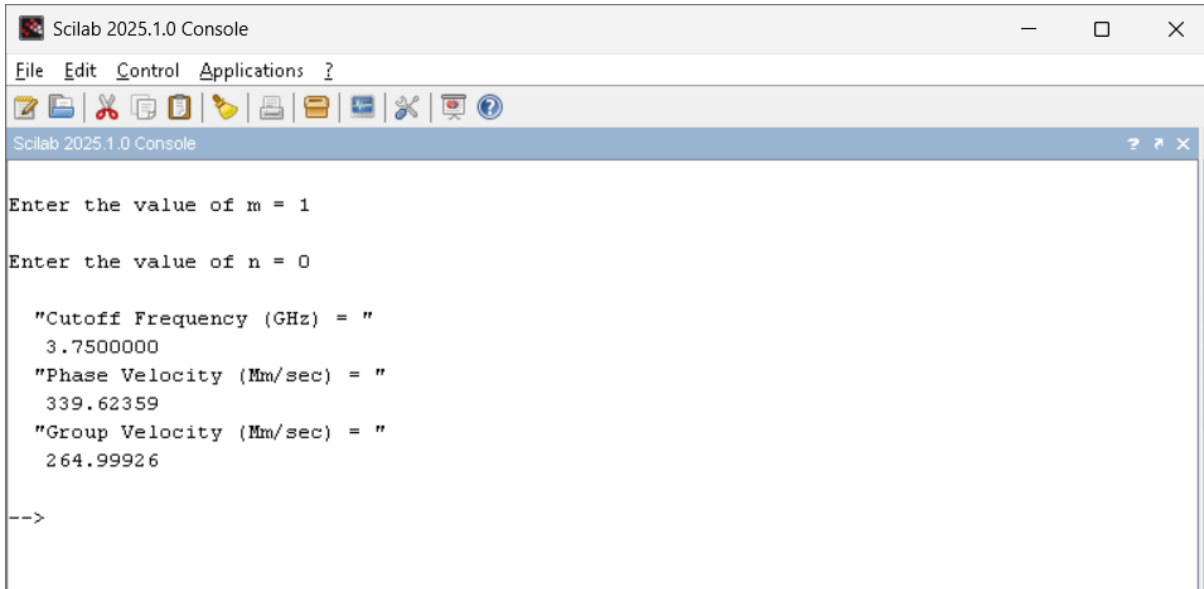
#### TEST CASE 1

#### PROGRAM :

```
clc;
clear;
close;

// Rectangular Waveguide Parameters
a = 4.0 * 10^-2; // 4 cm
b = 2.0 * 10^-2; // 2 cm
f = 8 * 10^9;    // 8 GHz
V = 3 * 10^8;    // Speed of light
m = input("Enter the value of m = ");
n = input("Enter the value of n = ");
// Cutoff Frequency
fc = (V/2) * sqrt((m/a)^2 + (n/b)^2);
disp("Cutoff Frequency (GHz) = ");
disp(fc * 10^-9);
// Phase Velocity
A = V / sqrt(1 - (fc/f)^2);
disp("Phase Velocity (Mm/sec) = ");
disp(A * 10^-6);
// Group Velocity
B = V^2 / A;
disp("Group Velocity (Mm/sec) = ");
disp(B * 10^-6);
```

## OUTPUT :



```
Scilab 2025.1.0 Console

Enter the value of m = 1

Enter the value of n = 0

"Cutoff Frequency (GHz) = "
3.7500000
"Phase Velocity (Mm/sec) = "
339.62359
"Group Velocity (Mm/sec) = "
264.99926

-->
```

## TEST CASE 2

### PROGRAM

```
clc;

clear;

close;

//TE10 Mode

a=8.636*10^-2;

b=4.318*10^-2;

f=4*10^9;

V=3*10^8;

m=input("enter the value of m= "); n=input("enter the value of n= ");

fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);

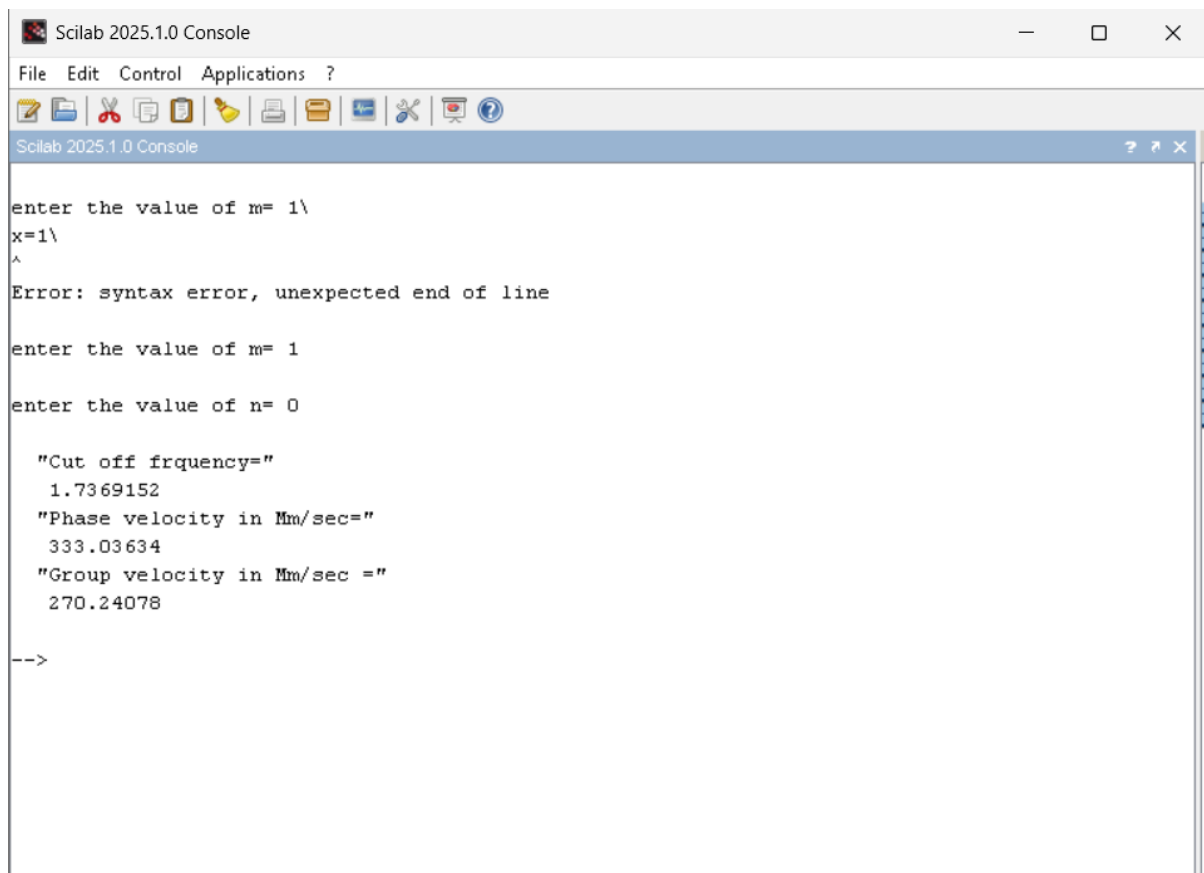
disp("Cut off frequency=",fc*10^-9);

//let phase velocity = A

A=V/sqrt(1-(fc/f)^2);
```

```
disp("Phase velocity in Mm/sec=",A*10^-6);  
  
//let group velocity = B  
  
B=V*V/A;  
  
disp("Group velocity in Mm/sec =", B*10^-6);
```

## OUTPUT



The image shows a screenshot of the Scilab 2025.1.0 Console window. The window has a title bar with the Scilab logo and the text "Scilab 2025.1.0 Console". Below the title bar is a menu bar with "File", "Edit", "Control", "Applications", and "?". Under the menu bar is a toolbar with various icons for file operations, editing, and help. The main area of the window is a text editor displaying the following text:

```
enter the value of m= 1\  
x=1\  
^  
Error: syntax error, unexpected end of line  
  
enter the value of m= 1  
  
enter the value of n= 0  
  
"Cut off frquency=" 1.7369152  
"Phase velocity in Mm/sec=" 333.03634  
"Group velocity in Mm/sec =" 270.24078  
  
-->
```

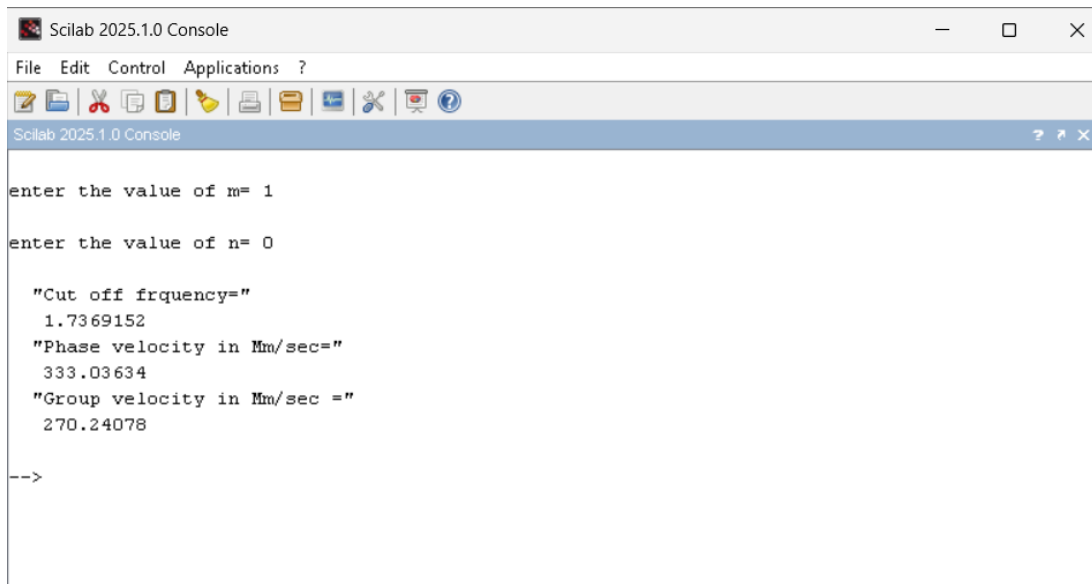
The text shows a script execution. It starts with a prompt "enter the value of m= 1\" followed by "x=1\" and a carriage return. This results in a syntax error: "Error: syntax error, unexpected end of line". After the error, there are two more prompts: "enter the value of m= 1" and "enter the value of n= 0". Following these prompts, the script outputs three lines of results: "Cut off frquency=" 1.7369152, "Phase velocity in Mm/sec=" 333.03634, and "Group velocity in Mm/sec =" 270.24078. The window ends with a "-->" prompt.

### TEST CASE 3

#### PROGRAM

```
clc;
clear;
close;
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m= ");
n=input("enter the value of n= ");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frequency=",fc*10^-9);
A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity = B
B=V*A/V;
disp("Group velocity in Mm/sec =", B*10^-6);
```

#### OUTPUT :



```
Scilab 2025.1.0 Console
File Edit Control Applications ?
enter the value of m= 1
enter the value of n= 0

"Cut off frquency="
1.7369152
"Phase velocity in Mm/sec="
333.03634
"Group velocity in Mm/sec ="
270.24078

-->
```

## TEST CASE 4

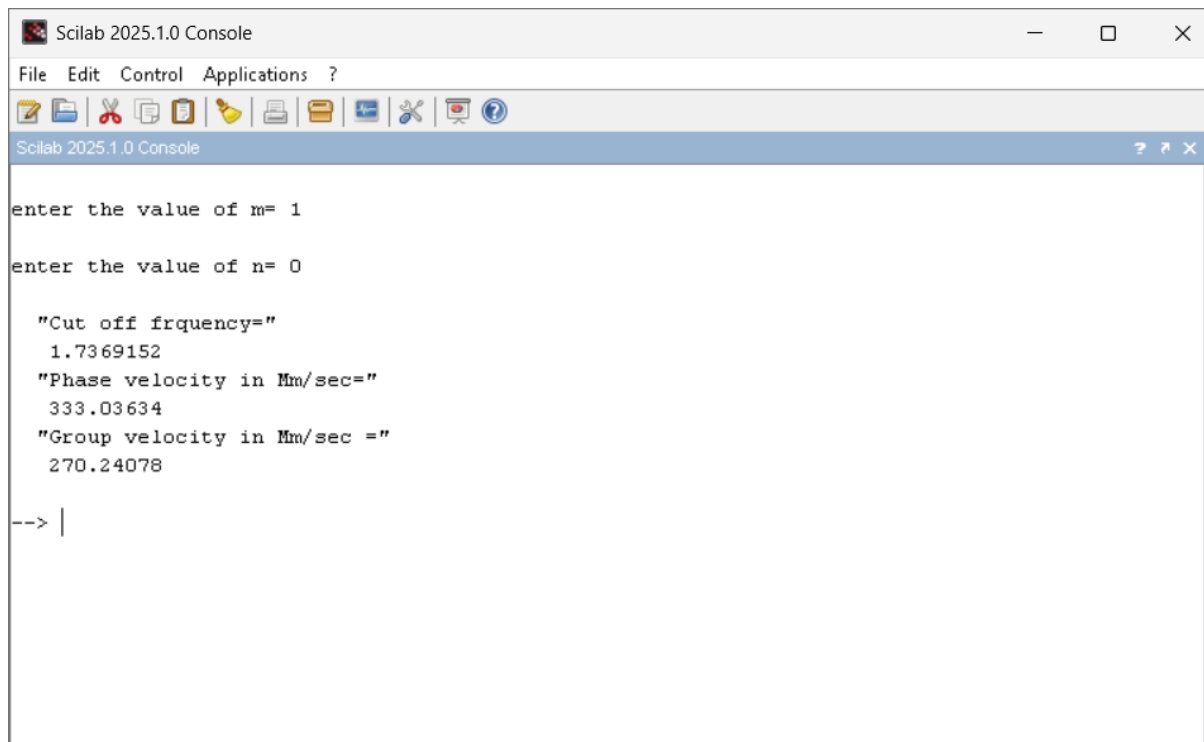
### PROGRAM

```
clc;
clear;
close;
//TE10 Mode
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m= ");
n=input("enter the value of n= ");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frquency=",fc*10^-9);
//let phase velocity = A
A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity = B
```

$B = V \cdot V / A;$

`disp("Group velocity in Mm/sec =", B*10^-6);`

## OUTPUT:



The image shows a screenshot of the Scilab 2025.1.0 Console window. The window has a title bar with the Scilab logo and the text "Scilab 2025.1.0 Console". Below the title bar is a menu bar with "File", "Edit", "Control", "Applications", and "?". Under the menu bar is a toolbar with various icons for file operations, editing, and help. The main area of the window is a text editor displaying the following text:

```
enter the value of m= 1
enter the value of n= 0

"Cut off frquency="
1.7369152
"Phase velocity in Mm/sec="
333.03634
"Group velocity in Mm/sec ="
270.24078

--> |
```